Paper 2

1 (a) (i) As the velocity of the ball increases from $t = 0$ to $t = 8.0$ s, the acceleration of the ball given by the gradient, $\frac{dv}{dt}$ decreases which implies that resultant force on the ball decreases.

The resultant force on the ball as it is falling from rest is given by the vector sum of the drag force F_D acting on the ball and its weight W i.e. $F_{resultant} = W - F_D$. Since W is constant, the drag force acting on the ball increases.

(ii) $$a = \frac{20.50 - 12.00}{6.0 - 1.0} = 1.70 \text{ m s}^{-2}$$

$$mg - F_D = ma$$
$$(0.80)(9.81) - F_D = (0.80)(1.70)$$
$$F_D = 6.488 = 6.5 \text{ N}$$

(b) (i) $$\eta = \frac{F_D}{6\pi rv}$$

$$\text{base units of } F_D = \text{kg} \times \left(\text{m s}^{-2}\right)$$

$$\text{base units of } \eta = \frac{\text{kg} \times \left(\text{m s}^{-2}\right)}{\text{m} \times \left(\text{m s}^{-1}\right)}$$
$$= \text{kg m}^{-1} \text{ s}^{-1}$$

(ii) The maximum drag force will be larger as it is always equal to the weight of the ball (at terminal velocity).